

9708/31/m/j/19

1-A:

A: An economy is considered Pareto efficient when it is impossible to make one person better off without making someone else worse off.

In other words, if there are resources or opportunities that can be reallocated to benefit some individuals without harming others, the economy is considered inefficient. This means that there is room for improvement, and resources can be better allocated to increase overall well-being.

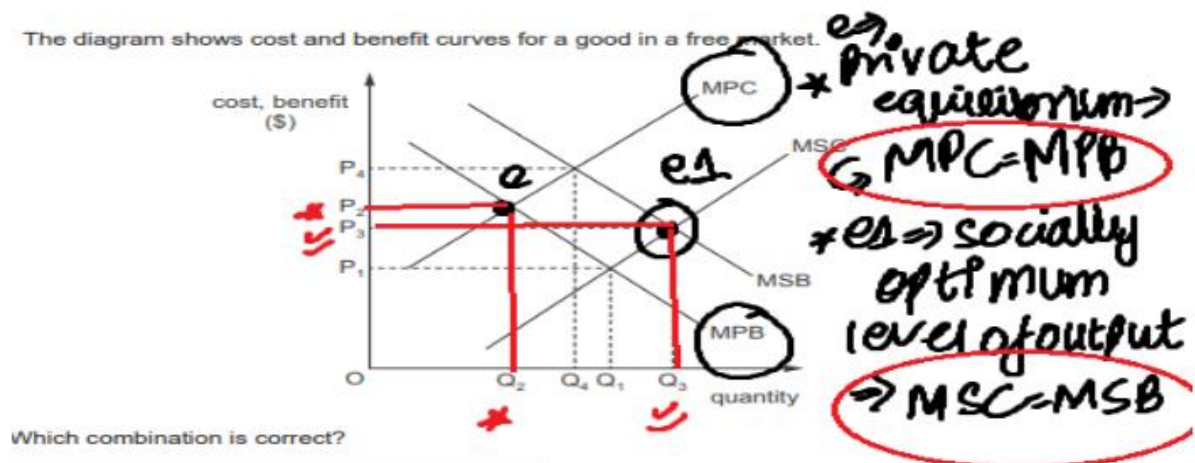
This is correct because it shows that the economy is operating inside its production possibility curve.

B: This option is incorrect because economic efficiency is primarily concerned with maximizing output with given resources, not necessarily with distribution of income.

C: This option is also incorrect because the size of the government sector versus the private sector does not necessarily determine economic efficiency. Economic efficiency is about optimal allocation of resources, regardless of whether it's the government or private sector doing the allocating.

D: This option is incorrect because when wages rise faster than production, this may lead to a rise in costs of production which can contribute to cost push inflation.

2-B:



3-C:

The difference between **social cost and private cost** is external cost, and we are interested in an option that reduces external cost.

A – If local residents are given the right to buy the firm's food at a discount, this would not help to reduce the external cost.

B – If the government subsidizes the food processing firm which discharges waste water, the output of the food processing firm will increase, and therefore, the negative externalities will increase.

C – When the lake is bought by the food processing firm, this means that the firm will now have "property rights". When the food processing firm becomes the owner of the lake, they have a direct financial stake in its well-being. With **property rights**, the firm is **accountable for any negative impacts its activities have on the lake's water quality**. This accountability can motivate the firm to implement measures **to reduce or eliminate the discharge of harmful wastewater**, as it directly affects their **property's value and reputation**.

D -When the waste water flows from the lake to the sea, this represents the existence of the "external costs"

4-D:

A: Option A is on a lower indifference curve which means that it will yield a lesser satisfaction.

B: Option B is on a lower indifference curve which means that it will yield a lesser satisfaction.

C: Option C represents the level of satisfaction that is currently un-attainable

D: The further the indifference curve is from the origin, the higher will be the satisfaction that it will yield . Therefore, D will be correct since it is on the highest indifference curve, which is currently attainable.

5-A:

A – In case of a giffen good, the substitution effect of a fall in price will be that it will result in an extension in qty demanded, which means that the change in demand will be "positive". Moreover, a fall in price results in a rise in real income and subsequently, there will be a contraction in qty demanded of giffen goods, which means that the change in demand from the income effect will be "negative".

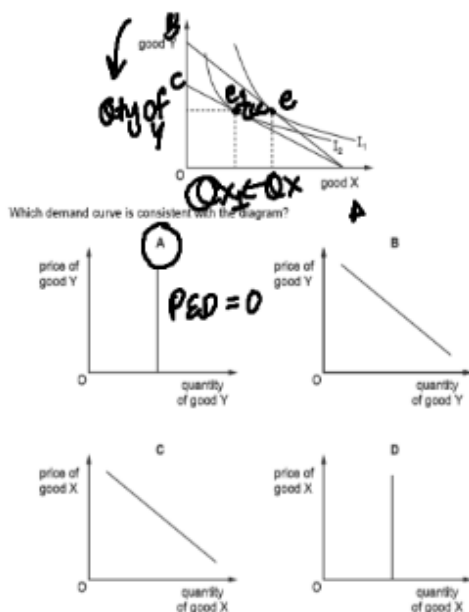
If for example, the price of a loaf of bread falls, people will contract some qty of loaves to purchase some meat which they can now afford, in order to feel richer.

B- In case of a giffen good, the substitution effect of a rise in price will be that it will result in a contraction in qty demanded, which means that the change in demand from substitution effect will be “negative”. And since a rise in price results in a fall in real income, it will result in an extension in qty demanded of giffen goods, which means that the demand change from the income effect will be “positive”

C – In case of inferior goods, the substitution effect of a fall in price will be an extension in qty demanded, which means that the change in demand from substitution effect will be “positive”. And since, a fall in price results in a rise in real income, it will result in a contraction in qty demanded of inferior good . This means that the change in demand from the income effect will be “negative”

D – In case of inferior goods, the substitution effect of a rise in price will cause a contraction in qty demanded, which means that the change in demand from the substitution effect will be “negative”. A rise in price will cause a fall in real incomes, and an extension in qty demanded of inferior goods, which means that the change in demand from the income effect will be “positive”

6-A:



- The budget line has tilted inwards and indifference curve is going inwards from I_1 to I_2 , which means that the price of Y has increased.

- And quantity of Y is unchanged even after a rise in price, which means that the demand is "perfectly inelastic"

- Price of X has not changed since we are at point A only, but qty of X is changing.

A - This graph shows that the demand for good Y is perfectly inelastic, which is why, this is correct.

B - This graph shows that the qty of good Y is changing, and the graph above doesnot represent this, which is why, this becomes incorrect.

C- This is incorrect as it shows that the price of X is changing, which is not the case here.

D - This is incorrect as it shows that the qty of good X is unchanged, which is not the case here.

7-D:

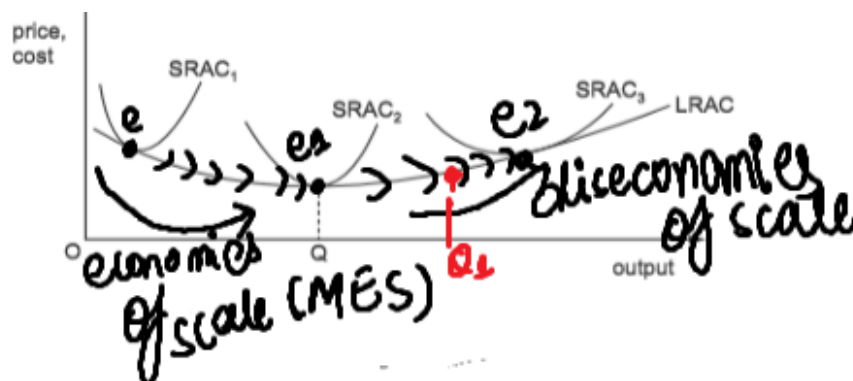
A: Small firms mainly provide a narrower product range rather than a “much wider range”, this is the reason why, they do not benefit from “risk-bearing economies of scale” unlike large firms. And, this is why, they are not able to survive.

B: A higher minimum efficient scale means that firms are required to produce a “larger volume of output”(or in other words, operate on a larger scale) to achieve the MES point where the long-run average costs are minimized, and maximum economies of scale are exploited. This would prevent small firms from surviving.

C: If they pay higher wages to their staff, their costs would increase, thereby leading to them finding it difficult to survive.

D: Since small firms can manage to provide a personalized-consumer service, which large firms cannot provide, they are able to survive along with large firms. This is why, key D, is correct.

8-B:



A – When each of the three SRAC curve is U-shaped, this represents “law of diminishing returns, since it applies in the short-run.

B_ When the LRAC curve is upward sloping beyond output OQ, as can be seen in the figure 1, that an increase in output from Q to Q₁, for example, would lead to a rise in long-run average costs, in turn representing “dis-economies of scale” or “decreasing returns to scale”. This is why, key B, becomes correct.

C- When the minimum point of SRAC 2 is below that of SRAC₁, this means that a movement from SRAC₁ to SRAC₂, would result in a fall in long-run average costs, or in other words, economies of scale.

D: When the minimum point of SRAC₂ is below that of SRAC₃, this means that a movement from SRAC₂ to SRAC₃, would result in diseconomies of scale.

9-C:

A: As the firm expands, it benefits from economies of scale and due to a larger volume of output being produced, energy costs can be distributed over a larger volume of output, thereby resulting in low energy costs per unit.

B: As the firm expands, the cost of components reduces since the firm is likely to obtain trade discounts on bulk-buying, which represents purchasing economies of scale

C: As the firm expands, there are managerial concerns known as “managerial diseconomies of scale”, and as a result of which, greater costs are incurred in monitoring workers. This is why, this option becomes correct.

D: The rate of interest paid on borrowing reduces because banks are likely to offer less interest rates to these firms in order to retain them, and also the administrative costs of processing these large loans is high, which is why, this option is incorrect.

10-B:

A – This is incorrect because in monopolistic competition, there is not a small number of buyers in the long run.

B – This is correct because unlike perfect competition in the case of which homogeneous products are produced, there is product differentiation in the long-run.

C – This is incorrect because both these market structures earn normal profits in the long run.

D – This is incorrect because none of them has barriers to entry in **the long-run**.

11-A:

Movement from X to Y leads to economies of scale being achieved and long run average costs to be minimized.

A – If the new airline can negotiate discounts when buying fuel, this represents “purchasing economies of scale”, as a result of which, long-run average costs are reduced. This is why, this option becomes correct.

B- If the airline has many layers of management, then it is likely to experience managerial diseconomies of scale, because of which the average costs are likely to increase.

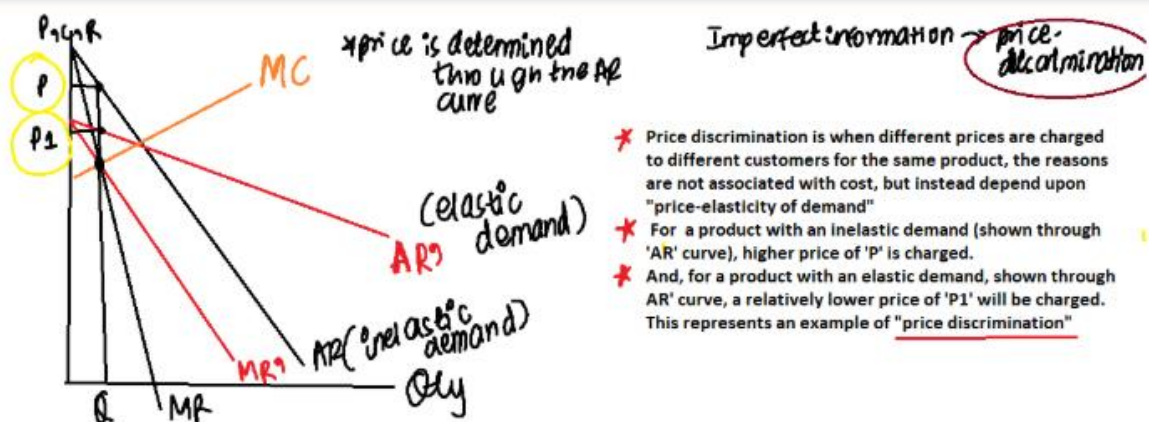
C- If there is a shortage of pilots, this means that higher wages will have to be paid, because of which “average total costs” are likely to rise.

D – If the airline’s workforce is demotivated, this represents labor dis-economies of scale, and because of lower productivity of workers, it is likely to experience higher average total costs.

12-C:

It can be seen below in the diagram, that for "price-discrimination" to be profitable, there should be different elasticities in different markets. The market that is more elastic(market M) with a PED of -1.8, represents a group of more-price sensitive customers, as compared to market N, that is relatively less-elastic(with a PED of -1.2).

Therefore, for market "M" relatively lower prices will have to be kept as compared to market "N"



Importantly, "the greater the geographical difference", the greater is the price difference that can be kept.

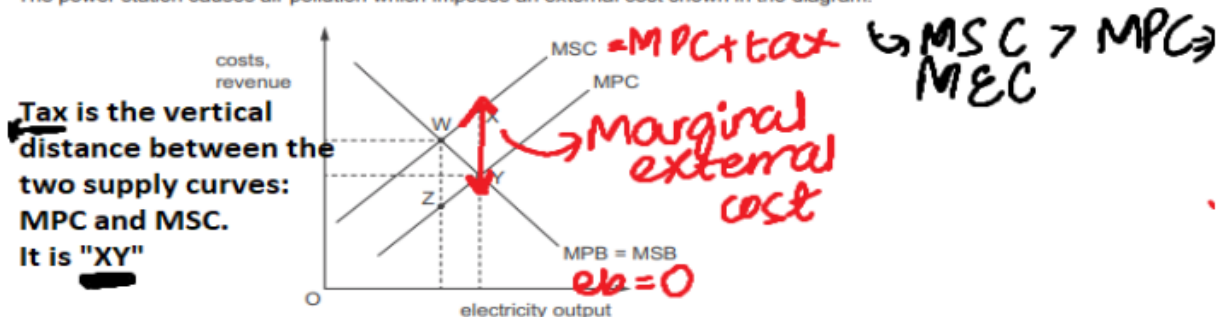
For example: airlines keep a price difference when serving "domestic markets" and "foreign markets".

This is why, key C, becomes correct.

13-C:

A community relies on a private company to supply electricity using a coal-fired power station.

The power station causes air pollution which imposes an external cost shown in the diagram.



The government decides to impose a tax on the electricity company to correct market failure.

What represents the tax?

A WX

B WY

C XY

D YZ

14-C:

The correct answer is C poverty trap.

A **poverty trap** refers to a situation where individuals are discouraged from working or increasing their income because it leads to a reduction in government benefits, leaving them no better off or even worse off. This is exactly what's happening in the scenario described: the workers increase their hours, but the resulting reduction in benefits means they don't see an increase in their overall income.

Here's a brief explanation of the other options:

A. Debt trap: a situation where individuals are unable to pay off their debts and are forced to take on more debt to service existing debts.

B. Liquidity trap: a situation in economics where people hoard cash and are unwilling to invest or spend, even at very low interest rates.

D. Unemployment trap does not describe the scenario given.

15-A:

A - Minimum wage laws aim to ensure a fair wage for workers, but **if they are set below the equilibrium market wage rate, they may be ineffective.**

Here's why the other options are more effective:

B: Progressive personal income taxes reduce income inequality by taxing higher earners at a higher rate since they take up a greater proportion of incomes of higher income groups, redistributing wealth to lower-income households through government programs.

C: Child benefits for low-income households represent mean tested benefits that are aimed to directly support vulnerable families, helping to reduce poverty and income inequality.

D: Removing sales taxes on food for lower-income households, which are regressive, makes essential items more affordable, increasing their purchasing power and since even after this, the qty demanded of essential items extends by a relatively smaller percentage since they are inelastic in nature, total expenditure of low income groups is most likely to fall.

16-D:

8

The table shows the main characteristics of employment in two occupations.

	occupation X	occupation Y
average annual wage	\$100 000	\$60 000
number of weeks annual leave	5 weeks	10 weeks
average length of working week	48 hours	44 hours
job security	low	high
length of training course to obtain job qualification	1 year	2 years

A – Although workers employed in occupation Y are given a “higher degree of job security”, it does not mean that the workers attach a “greater importance to it”. Based on this, we cannot get to know about the individual preferences of workers working in the industry. **It may be that they attach a higher importance to wages.**

B – Although, workers employed in occupation X have a working week of 48 hours, representing that they have a higher number of working hours as compared to occupation Y, this does not mean that they attach lesser importance to leisure activities. They are not even getting high job security.

C – We don’t know which job is popular among workers, because that will depend on individual preferences of workers. This is important because it will determine the competition for training places to enter occupation X or Y.

D – This is correct to say because occupation Y offers more non-pecuniary or nonfinancial advantages as compared to occupation X. This is because, it offers a higher duration of annual leave in weeks, and also, the average length of working week is relatively lesser, and importantly, there is a higher degree of job-security available.

17-B:

The table shows a firm that has four possible methods of production to make its desired output. Each method has a different cost and causes a different amount of pollution.

The government rules that polluting firms must buy pollution permits that cost \$90 each. Each permit allows the firm to emit 1 tonne of pollution. No pollution can be emitted without a permit.

Which method of production has least total cost?

	cost of production (\$ per day)		pollution caused (tonnes per day)			
A	320	+	3	$\times 90 = 270$	590	
B	380	+	2	$\times 90 = 180$	560	least total cost
C	480	+	1	$\times 90 = 90$	570	
D	700	+	0	$\times 90 = 0$	700	

This is why, key B, becomes correct.

18-B:

GDP at market prices = GDP at factor cost + taxes- subsidies.

19-D:

Together, these factors suggest that GDP calculations are relatively accurate because:

- The population is stable, so changes in GDP reflect changes in economic activity, not population growth.
- The informal economy is small, so most economic activity is captured in GDP estimates.

- Stable inflation means that price changes don't distort GDP calculations.

As a result, GDP can provide a relatively accurate picture of changes in the standard of living over time.

20-A:

-> Based on the data, the country most likely to be a developed country is A, with a primary sector share of 15%, secondary sector share of 40%, and tertiary sector share of 45%.

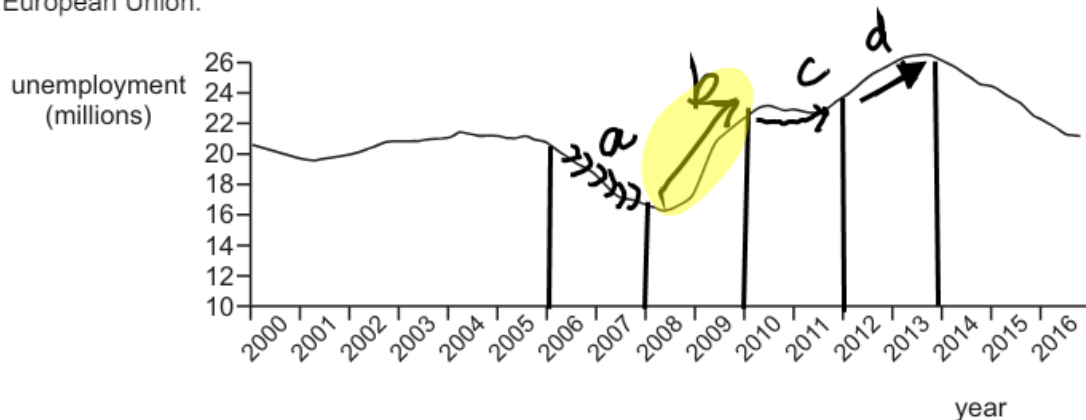
In developed countries, the tertiary sector (services) typically dominates the economy, while the primary sector (agriculture) is relatively small. This is because developed countries have largely transitioned from agricultural to service-based economies.

Country A's sector distribution, with a strong tertiary sector and a relatively small primary sector, suggests a developed economy.

21-B:

It can be seen in the graph that the highest rate of **an increase in unemployment** was experienced in between **2008 to 2010**. **This is because, the curve is the steepest there.**

21 The diagram shows unemployed persons (in millions) for the years 2000 to 2016 for the European Union.



During which period was the highest rate of increase in unemployment experienced?

- A 2006 to 2008 *a*
- B 2008 to 2010** *b*
- C 2010 to 2012 *c*
- D 2012 to 2014 *d*

22-C:

The correct answer is C structural.

The government is trying to reduce structural unemployment, which occurs when workers' skills become obsolete due to changes in the pattern of demand and supply in the market, especially when they are specialized, which makes them occupationally mobile. **In this**

case, the coal mine closure is a result of a structural shift in the energy market, as consumers switch to cleaner fuels.

By providing retraining courses and recruitment events, the government is helping workers adapt to this structural change and find new employment opportunities in different industries or sectors.

A. Cyclical unemployment: related to a downturn in the business cycle, characterized by **too low of Aggregate demand**, when there are job losses across a range of industries.

B. Seasonal unemployment: occurs in industries with seasonal fluctuations, like agriculture or tourism.

D. Technological unemployment: specifically refers to job losses due to automation or technological advancements.

23-B:

A – Self sufficiency can lead to missed opportunities for trade

B. Market access is crucial for developing countries to expand their exports and drive economic growth. By gaining access to developed country markets, they can increase their foreign exchange earnings and utilize their comparative advantage. This is in line with **trade not aid policy**.

C. The "trade not aid" approach emphasizes market access over aid.

D. Import substitution policies aim to promote domestic production by erecting trade barriers. However, this approach can limit access to foreign markets and subsequently opportunities to trade internationally may be missed.

24-C: Method 1:

24 The table shows the values of selected macroeconomic variables over a period.

* formula of 4 sector multiplier

$$\frac{1}{m_p s + m_p t + m_p m}$$

	year 1 (\$m)	year 2 (\$m)	
investment	200	220	
saving s	180	190	$\Delta 10$
exports	100	110	
imports m	120	140	$\Delta 20$
government expenditure	150	160	
taxation t	150	160	$\Delta 10$
equilibrium national income	1800	2000	$\Delta 200$

$$\frac{1}{0.05 + 0.1 + 0.05} = \frac{1}{0.2} = 5$$

What is the value of the multiplier?

A 2

B 4

☒ C 5

D 10

$$m_p s = \frac{\Delta s}{\Delta y} \Rightarrow \frac{10}{200} \Rightarrow 0.05$$

$$m_p m = \frac{\Delta m}{\Delta y} \Rightarrow \frac{20}{200} \Rightarrow 0.1$$

$$m_p t = \frac{\Delta t}{\Delta y} \Rightarrow \frac{10}{200} \Rightarrow 0.05$$

Method 2:-

Autonomous expenditures
I, X, G

	year 1 (\$m)	year 2 (\$m)
investment	200	220
saving	180	190
exports	100	110
imports	120	140
government expenditure	150	160
taxation	150	160
equilibrium national income	1800	2000

What is the value of the multiplier?

A 2

B 4

C 5

D 10

$$\frac{\text{change in RNI}}{\text{change in AE}} \Rightarrow \frac{200}{20 + 10 + 10}$$

$$\Delta I \quad \Delta X \quad \Delta G$$

$$\Rightarrow \frac{200}{40} \Rightarrow \text{multiplier} \Rightarrow \textcircled{5}$$

25-D:

A. Business confidence is likely to decrease in an economy with a negative output gap. When actual output is lower than potential output, businesses may become cautious and hesitant to invest or expand, leading to reduced confidence in the economy.

B. The economic growth rate is likely to decrease in an economy with a negative output gap. A negative output gap indicates underutilization of resources **arising from too low of Aggregate demand.**

C - The inflation rate is likely to decrease in an economy with a negative output gap. With reduced Aggregate demand relative to Aggregate supply, there is likely to be surplus of resources in an economy, and this deflationary gap will lead to a fall in the general price level of an economy

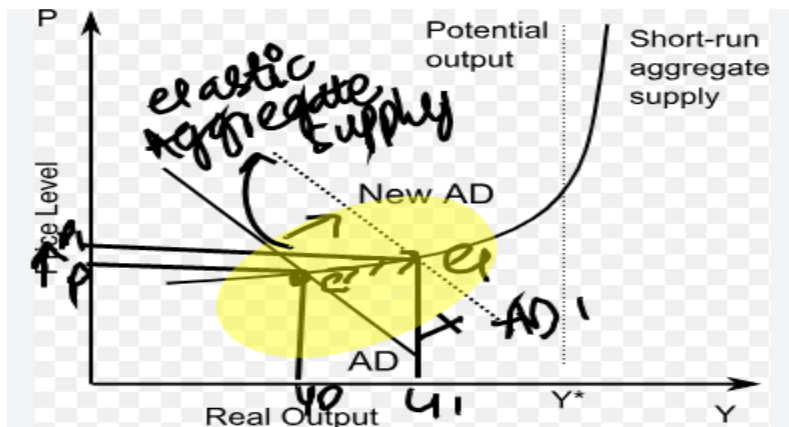
D. The unemployment rate is likely to increase in an economy with a negative output gap. As businesses produce less, they may lay off workers, leading to higher unemployment. Since negative output gap represents too low of Aggregate demand, there is likely to be an **increase in demand deficient unemployment** in this case.

26-C:

An expansionary fiscal policy involves cutting taxes and/ or increasing government spending in an economy.

If because of a fall in income taxes, disposable incomes rise, this will stimulate an increase in consumption expenditure, thereby boosting Aggregate demand.

- **Elastic aggregate supply curve:** This means that increases in Aggregate demand will have a greater expansionary effect on real incomes as they will increase by a greater percentage than an increase in prices indicating that there would not be "too much of inflation"



Wage flexibility: low

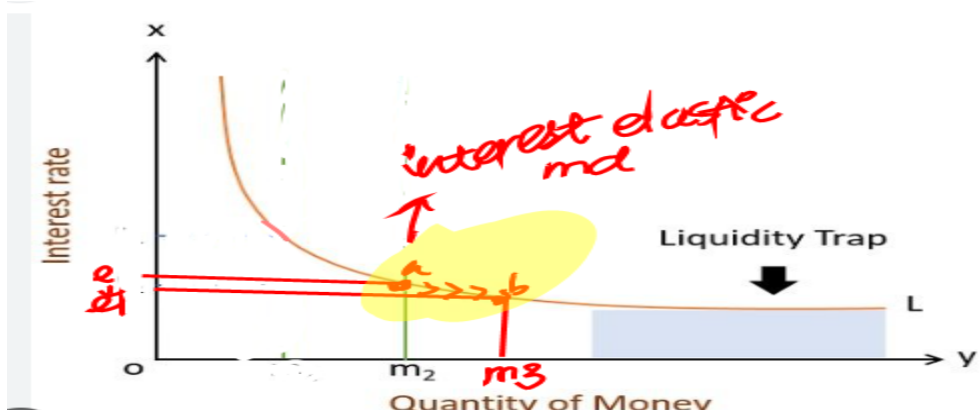
With low wage flexibility, an expansionary fiscal policy is more likely to be effective without causing excessive inflation because:

- **Wages won't chase prices:** If prices rise due to increased demand, wages won't immediately follow suit, reducing the risk of a wage-price spiral. The slow adjustment of wages will prevent a rapid increase in prices, containing inflationary pressures.

Demand for money: Interest elastic:

When a government increases its spending by borrowing from the central bank, this will increase money supply. In this case, since the demand for money is interest elastic, the increase in money supply will have a minimal effect on interest rates. Since interest rates don't change much, inflation (the rate at which prices rise) is also less likely to increase.

This can be seen in the diagram below:



27-B:

A – A fall in the price of bonds represents that the interest rates are high since there is a negative relationship between the price of bonds and interest rates. This means that individuals would be incentivized to save more, because of which, when consumption falls, the transactionary demand for money will decrease too.

B – This is correct because an increase in income is likely to encourage an increase in the transactionary demand of money.

C –When interest rates are high, this means that individuals would be incentivized to save more, because of which, when consumption falls, the transactionary demand for money will decrease too.

D – An increase in unemployment indicates that purchasing powers of individuals will fall, which is likely to bring about a decrease in the transactionary demand of money.

28-B:

A. This would actually increase bank lending, as banks would have more funds to lend. When commercial banks have more balances with the central bank, it means they have more reserves and are more likely to lend.

B. This would lead to a reduction in bank lending. Liquidity ratios require banks to hold a certain percentage of their deposits in reserve, rather than lending them out. If liquidity ratios increase, banks must hold more in reserve and lend less.

C. This would also increase bank lending. A monetary base is **the total amount of a currency in general circulation or in the commercial bank** deposits held in the central bank's reserves. An increase in the monetary base means more funds are available for banks to lend.

D. This would also increase bank lending. Liquid assets are funds that banks can easily use to lend. If banks have more liquid assets, they have more funds available to lend.

29-C:

- 29 The table gives details of the inflation and unemployment rates for different countries in 2008 and 2011.

country	inflation rate (% per annum)		unemployment rate (% of workforce)	
	2008	2011	2008	2011
Canada	2.38	2.89	6.15	7.47
France	3.16	2.29	7.81	9.68
Germany	2.75	2.48	7.60	5.98
Japan	1.37	-0.28	3.99	4.55

Which country or countries experienced a trade-off between their inflation and unemployment rates?

- A Canada and Germany
- B Canada only
- C France and Japan
- D France only

-ve relationship

30-A:

A - Raising the rate of Value Added Tax (VAT) or General Sales Tax suggests that the government's main macroeconomic objective is to increase tax revenue and reduce the budget deficit. This is because a higher VAT rate increases the amount of tax collected on each sale, generating more revenue for the government.

B: VAT will exacerbate income inequalities. This is because it is a regressive tax which means that it takes up a greater proportion of incomes of the lower income groups.

C: Raising VAT is unlikely to reduce unemployment. This is because it will increase the price of goods and services, as a result of which, real incomes of individuals and subsequently consumption will fall. This will further decrease Aggregate demand, as a result of which, there is likely to be an increase in unemployment.

D . After having raised the costs of production, VAT will reduce the Aggregate supply of the economy, potentially **contributing to cost push inflationary pressures**.

- Low wage flexibility: This means that wages are sticky and don't adjust quickly to changes in prices, reducing the risk of wage-price spirals and inflation.

